

LECTURE 6

Topics

Equilibrium concept in capital markets, the market portfolio, the Capital Asset Pricing Model (CAPM), the relationship between risk premium and market betas in capital markets, CAPM restriction on alphas, testing the CAPM, extensions to non-market sources of systematic risk, Arbitrage pricing theory (APT), multi-factor asset pricing model of Fama and French, the size, and value factors

- Recall the main results of the optimal portfolio analysis:
 - The separation result: all investors choose to invest in the same two assets: the optimal risky portfolio (or the tangency portfolio, R) and the risk-free asset, F
 - An investor's complete portfolio (the mix of T and F) depends on the investor's risk aversion coefficient A :
 - Weight of optimal risky portfolio: $y = \frac{E(r_T) - r_F}{\sigma_T^2 \times A}$
 - Weight of the risk-free asset: $1 - y$
- Suppose the weight of asset i in the tangency portfolio T is w_i

- Consider now an investor k with capital z_k
- Then, investor k 's dollar investment in asset i becomes:

$$z_k \times y_k \times w_i = \frac{E(r_T) - r_F}{\sigma_T^2} \times w_i \times \frac{z_k}{A_k}$$

- This is the demand for asset i from investor k
- If we sum up this demands across all investors we obtain the aggregate demand for asset i :

$$D_i = \frac{E(r_T) - r_F}{\sigma_T^2} \times w_i \times \sum_k \frac{z_k}{A_k}$$

- If the price of asset i is p_i and if there are N_i units of asset i available on the market, then the aggregate supply of asset i is:

$$S_i = p_i \times N_i$$

- Because asset i trades on a capital market, the price p_i adjusts such that *aggregate demand = aggregate supply*

$$D_i = S_i$$

- When aggregate demand = aggregate supply we say that the market for asset i is in *equilibrium*
- Thus, when the market for asset i is in equilibrium we have:

$$\frac{E(r_T) - r_F}{\sigma_T^2} \times w_i \times \sum_k \frac{z_k}{A_k} = p_i \times N_i$$

- By extension, if the markets for all assets j are in equilibrium, then the previous equation holds for any asset j as well (simply replace i with j)

- When capital markets are in equilibrium (market for any asset j is in equilibrium), then we can sum across j in the previous equation:

$$\frac{E(r_T) - r_F}{\sigma_T^2} \times \sum_k \frac{Z_k}{A_k} \times \sum_j w_j = \sum_j p_j \times N_j$$

- But $\sum_j w_j = 1$ because the weights of the tangency portfolio add up to 1
- Thus, we obtain that:

$$\frac{E(r_T) - r_F}{\sigma_T^2} \times \sum_k \frac{Z_k}{A_k} = \sum_j p_j \times N_j$$

- Substituting this formula back into the equilibrium equation for asset i , we obtain:

$$w_i \times \sum_j p_j \times N_j = p_i \times N_i$$

- Rearranging the previous formula, we obtain a very interesting result

Equilibrium weights for the optimal risky portfolio

When capital markets are in equilibrium, asset prices adjust such that the weight of asset i in the optimal risky portfolio is given by the ratio of the market capitalization of asset i to the total market capitalization:

$$w_i = \frac{p_i \times N_i}{\sum_j p_j \times N_j}$$

- This result is saying that when capital markets are in equilibrium, the optimal risky portfolio becomes the value-weighted portfolio of all assets in the market
- In other words, in equilibrium, the optimal risky portfolio looks like S&P 500

- The fact that in equilibrium asset prices adjust until demand equals supply means that asset prices have to behave in a certain way
- To understand how this works, we first recall that the tangency portfolio T has the largest Sharpe's ratio

$$SR_T = \frac{E(r_T) - r_F}{\sigma_T}$$

- In particular, if we were to invest a in asset i and $1 - a$ in T , then the resulting portfolio, P , can never improve on the Sharpe's ratio of T :

$$\max_a SR_P = SR_T$$

- Next we show how this equation turns into a constraint for prices

- The risk premium of portfolio P can be calculated with the formula:

$$E(r_P) - r_F = a \times (E(r_i) - E(r_T)) + E(r_T) - r_F$$

- While the return variance of portfolio P becomes:

$$\sigma_P^2 = a^2 \sigma_i^2 + (1 - a)^2 \sigma_T^2 + 2 \times a \times (1 - a) \times COV(r_i, r_T)$$

- And the Sharpe's ratio of portfolio P becomes:

$$SR_P = \frac{a \times (E(r_i) - E(r_T)) + E(r_T) - r_F}{\sqrt{a^2 \sigma_i^2 + (1 - a)^2 \sigma_T^2 + 2 \times a \times (1 - a) \times COV(r_i, r_T)}}$$

- Recall now from your managerial economics class that we can find the weight a that maximizes the Sharpe's ratio of portfolio P by taking a derivative and setting it equal to zero

- We already know that $a = 0$ is the weight that maximizes the Sharpe ratio of portfolio P
- Thus, the derivative of the Sharpe's ratio evaluated at $a = 0$ must equal zero
- This constraint can be formalized as follows:

$$\frac{E(r_i) - E(r_T)}{E(r_T) - r_F} = \frac{COV(r_i, r_T) - \sigma_T^2}{\sigma_T^2}$$

- This ratio is equivalent to a simpler one:

$$\frac{E(r_i) - r_F}{E(r_T) - r_F} = \frac{COV(r_i, r_T)}{\sigma_T^2}$$

- The right-hand side of this equation is the *beta* of asset i :

$$\beta_i = \frac{COV(r_i, r_T)}{\sigma_T^2}$$

- Rearranging the previous equation, we obtain the main equilibrium asset pricing restriction:

$$E(r_i) - r_F = \beta_i(E(r_T) - r_F)$$

- This equation is saying that when capital markets are in equilibrium, the risk premium of asset i is proportional to the risk premium of the tangency portfolio T , by a degree commensurate with the beta of the asset, β_i
- This result together with the result about the weights of the tangency portfolio are the basis for the most important asset pricing model in finance: the *Capital Asset Pricing Model* or the CAPM

- We state the main implications of the CAPM in a result:

The CAPM

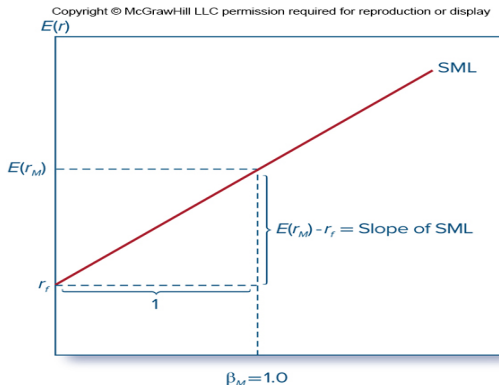
When capital market are in equilibrium, asset prices adjust such that the optimal risky portfolio (or the tangency portfolio) becomes the value-weighted portfolio of all the assets. As such, we call this optimal risky portfolio, the *Market* portfolio.

Furthermore, in equilibrium, asset prices adjusts such that the risk premium of an asset i is proportional to the risk premium of the market portfolio, by a degree commensurate with the market beta of the stock, β_i :

$$E(r_i) - r_F = \beta_i(E(r_M) - r_F)$$

- The main message of the CAPM is that the only type of risk that is *priced* by capital markets that are in equilibrium is *market risk*
- As a result, the CAPM suggests that the appropriate measure of risk is the market beta (i.e., market risk), β and not the return volatility (i.e., total risk), σ
- There some important key differences between β and σ :
 - σ is always positive, while β can be negative
 - An asset with a $\sigma = 0$ is effectively risk-free
 - However, an asset with a $\beta = 0$ can still be risky (all risk is going to be non-market risk)
 - The beta of the market portfolio always equals 1: $\beta_M = 1$

- The CAPM equation $E(r_i) - r_F = \beta_i(E(r_M) - r_F)$ suggests that when plotting expected returns, $E(r_i)$, against market betas, β_i , all assets should be on the line connecting the risk-free asset, F , and the market portfolio, M
- This line is called the *security market line* (SML)



- Of course, when markets are not in equilibrium, assets don't need to line up on the SML
- The deviation away from the SML is known as the *alpha* and is calculated with the formula:

$$\alpha_i = E(r_i) - r_F - \beta_i(E(r_M) - r_F)$$

- If the CAPM model is true, then assets can have $\alpha \neq 0$ if capital markets are not in equilibrium, but $\alpha = 0$, when capital markets reach their equilibrium

- Furthermore, if the CAPM is true, then:
 - $\alpha_i > 0$ means that asset i is currently *undervalued*, and its price is expected to go up (and as a result α_i approach 0)
 - $\alpha_i < 0$ means that asset i is currently *overvalued*, and its price is expected to go down (and as a result α_i approach 0)
 - $\alpha_i = 0$ means that asset i is currently fairly priced, and that the price is not expected to go neither up nor down (i.e., α_i remains at zero)

- We derived the CAPM under somewhat restrictive assumptions:
 - Investors focus on achieving short-term happiness (myopic) rather than life-time happiness (recall Lecture 1)
 - All available information on the assets is public, and all investors process it the same way (i.e., there is no private information or "skill" from extracting better quality signals from available information)
 - Investors don't care about transaction costs from rebalancing their portfolios and taxes owed on capital gains
- Most of these assumptions can be relaxed, but they will alter the CAPM and the main asset pricing restriction

- Relaxing these assumptions will change the CAPM in the same way: investors start to care about non-market risks that affect life-time happiness (rather than short-term happiness) and demand to be compensated for it
- Some of these non-market risks include:
 - Interest rate risk
 - Labor income risk
 - Liquidity risk (from costs associated with closing out a position in an asset)
- As a result of these changes, the main asset pricing implication of CAPM changes as well to reflect that investors demand compensation for both market and non-market risks:

$$E(r_i) - r_F = \beta_i^M(E(r_M) - r_F) + \beta_i^{NM}(E(r_{NM}) - r_F),$$

where $E(r_{NM}) - r_F$ is compensation for non-market risk

- The take-away from relaxing the main assumptions that support the CAPM is that once investors start caring about non-market risk they make very different choices for their optimal portfolio allocation
- For example, an entrepreneur has most of its capital invested in a business, which is a risky real asset that doesn't trade
- Naturally, the entrepreneur might be reluctant to take on even more risk by investing in risky financial assets
- Thus, entrepreneurs might under-invest in risky financial assets relative to non-entrepreneur investors
- Similarly, the trade-off between human capital (earning potential) and financial capital can also distort an investor's portfolio allocation decision

- We now focus on some practical issues of the CAPM
- Measuring the market portfolio
 - The CAPM predicts that the market portfolio should include all risky assets, tradeable or non-tradeable
 - Such a portfolio cannot be practically implemented and thus we need to approximate it with a portfolio that captures in spirit the idea of CAPM's market portfolio

- Measuring the market beta

- Once we decided on the proxy for the market portfolio, M^0 , we can calculate the market beta of an asset
- The market beta of asset i can be calculated directly with:

$$\beta_i = \frac{\text{Cov}(r_i, r_{M^0})}{\sigma^2(r_{M^0})}$$

- The beta can also be calculated from a regression:

$$r_i(t) - r_F = \alpha_i + \beta_i(r_{M^0}(t) - r_F) + e_i(t)$$

- Calculating the alpha
 - We can calculate alpha in two ways
 - If we know the asset's beta, β_i , we can calculate the asset's alpha as follows:

$$\alpha_i = E(r_i) - r_F - \beta_i(E(r_{M^0}) - r_F)$$

- Alternatively, we can calculate alpha from a linear regression:

$$r_i(t) - r_F = \alpha_i + \beta_i(r_{M^0}(t) - r_F) + e_i(t)$$

- Calculating the beta and the alpha of a portfolio of assets
- Suppose a portfolio P invests w_1 in asset 1, w_2 in asset 2, ..., and w_N in asset N
- Then the beta of portfolio P is:

$$\beta_P = w_1 \times \beta_1 + w_2 \times \beta_2 + \dots + w_N \times \beta_N$$

- The alpha of portfolio P is:

$$\alpha_P = w_1 \times \alpha_1 + w_2 \times \alpha_2 + \dots + w_N \times \alpha_N$$

- And as a reminder, the expected return of portfolio P is:

$$E(r_P) = w_1 \times E(r_1) + w_2 \times E(r_2) + \dots + w_N \times E(r_N)$$

- And the risk premium of portfolio P is

$$E(r_P) - r_F$$

- The CAPM is embraced by practitioners because of its simplicity
- Asset managers use it to estimate the exposure of their portfolios to market risk
- A common practice among asset managers is to neutralize the market risk of a portfolio by using S&P 500 futures in proportion to the portfolio's market beta
- Corporations use the CAPM in capital budgeting to estimate the cost of capital of a project
- If the beta of the project's cash flows is known, say β_{Proj} , then the project's cost of capital, $E(r_{Proj})$ can be calculated using the CAPM formula:

$$E(r_{Proj}) = r_F + \beta_{Proj}(E(r_{M^0}) - r_F)$$

- This costs of capital can then be used to perform discounted-cash flow (DCF) analysis and calculate the project's net present value (NPV)

- Of course, the CAPM is just a theoretical model that has no value if it has no support in the data
- However, testing the CAPM is complicated by the fact that we do not observe directly the CAPM's market portfolio
- In particular, if we use a proxy for this market portfolio (e.g., S&P 500), we must take any test results with a grain of salt
- The various tests of the CAPM center around the CAPM's asset pricing restriction that firms with higher betas should earn higher expected returns
- This CAPM restriction seems to be satisfied for many test assets, but not for all

- For example, if we sort stocks by size (market capitalization) and distribute them into size deciles, the 10 size portfolios seem to exhibit a clear trend in expected returns (higher for smaller deciles), but no trend in the portfolios' betas
- Similarly, if we sort stocks by book-to-market ratio (book equity to market capitalization) and distribute them into book-to-market (or value) deciles, the 10 value portfolios seem to exhibit a clear trend in expected returns (higher for higher value deciles), but no trend in the portfolios' betas
- These results were first documented by Fama and French in 1993, and they were viewed as "anomalies" relative to the CAPM
- To alleviate the concerns about the CAPM's market portfolio not being observable, Fama and French created their own market portfolio from all tradeable stocks in the US (this portfolio is the value-weighted CRSP portfolio in WRDS)

- The findings of Fama and French reinforced the view that the CAPM fails in certain tests because it doesn't take into account investors' concerns with non-market risk
- As a result, the finance academics started to look for more general versions of the CAPM that could also capture the non-market risk component and that could show better test results
- Two such models were the *Arbitrage Pricing Theory* (APT) and the *Fama-French 3-factor Model* (FF3M)
- Unlike the CAPM that is built from economic first principles (i.e., investors' preferences over risk and risk premium), the APT and the FF3M are exploiting price differences across similar assets

- The APT is build around the idea that capital markets are sufficiently efficient that no arbitrage opportunities exist at any given moment
- First recall the concept of arbitrage opportunity: an investment opportunity that generated non-negative profits, risk-free
- In capital markets, arbitrage opportunities arise when two assets that have identical cash flows trade for different prices
- Smart investors take advantage of these opportunities by short-selling the more expensive asset and buying the less expensive asset to pocket the difference in prices across the two assets
- For example, an arbitrage opportunity may arise if a 10-year US Treasury bond with one year to maturity and a 3-year US Treasury bond with one year to maturity trade for different prices

- We now apply this arbitrage concept to well-diversified portfolios
- Recall that in a well diversified portfolio, non-diversifiable or systematic risk is the only type of risk that matters
- Suppose further that the systematic risk is perfectly captured by a market portfolio, M
- Then, the return of a portfolio P can be decompose into a systematic component and a non-systematic component:

$$r_P(t) = E(r_P) + \underbrace{\beta_P \{r_M(t) - E(r_M)\}}_{\text{systematic risk}} + \underbrace{e_P(t)}_{\text{non-systematic risk}}$$

where $E(e_P) = 0$

- When portfolio P is well diversified, the non-systematic component, e_P , can be ignored:

$$r_P(t) = E(r_P) + \beta_P \{r_M(t) - E(r_M)\}$$

- We now argue that two well-diversified portfolios that have the same beta, must have the same expected returns
- To see this, consider two well-diversified portfolios, P and Q , that have the same beta $\beta_P = \beta_Q = 0.5$ but different expected annual returns: $E(r_P) = 8\%$ while $E(r_Q) = 10\%$.
- Suppose that you short-sell \$1 million worth of portfolio P to get \$1 million now, which you then use to purchase portfolio Q
- Then, the cash flows (CF) of your strategy are as follows:
 - CF Now: +\$1 million – \$1 million = \$0
 - CF in one year:

$$\begin{aligned} & -\$1\text{million} \times (1 + 8\% + 0.5 \{r_M - E(r_M)\}) \\ & +\$1\text{million} \times (1 + 10\% + 0.5 \{r_M - E(r_M)\}) \\ \hline & +\$1\text{million} \times (10\% - 8\%) = \$20,000 \end{aligned}$$

- This is an arbitrage opportunity that generates \$20,000 risk-free

- The example above illustrates that when there are no arbitrage opportunities, there has to be the case that two well-diversified portfolios with same beta must have the same expected return
- We can make this argument a bit more general
- The systematic risk could also come from non-market factors (NM) such as interest rates, oil shocks, etc
- The previous model for well-diversified portfolios becomes:

$$r_P(t) = E(r_P) + \beta_P^M \{r_M(t) - E(r_M)\} + \beta_P^{NM} \{r_{NM}(t) - E(r_{NM})\}$$

- The result above can be extended to this more general case: In the absence of arbitrage opportunities, well-diversified portfolios with same market and non-market betas, must have the same expected returns

- The results so far focus on well-diversified portfolios
- Can we extend them to non-diversified portfolios including individual assets?
- The answer is yes, for *almost* all individual assets
- These results collectively are known as the Arbitrage Pricing Theory:

The Arbitrage Pricing Theory

In the absence of arbitrage opportunities, *almost any* two individual assets with same market and non-market betas must have the same expected returns.

Furthermore, *any* two well-diversified portfolios with same market and non-market betas must have the same expected returns.

- As discussed above, a proper test of the CAPM is hard, if not impossible to set up, given that the market portfolio that the CAPM requires is not observable in the data
- However, even when construct our own versions of the market portfolio that are closer to what the CAPM requires (e.g., WRDS's CRSP value-weighted market portfolio), the CAPM seems to fail completely certain tests
- For example, if we sort stocks by size and distribute them into size deciles, the 10 size portfolios seem to exhibit a clear trend in expected returns, but no trend in the portfolios' betas
- Similarly, if we sort stocks by book-to-market ratio and distribute them into book-to-market (or value) deciles, the 10 value portfolios seem to exhibit a clear trend in expected returns, but no trend in the portfolios' betas

- Fama and French (Journal of Finance, 1993) argued that maybe the reason the CAPM fails to price the size and the value portfolios is because there are risk factors that are not picked up well by the market portfolio (a *hypothesis* that Fama and French wanted to test)
- To make their point, FF reverse-engineered two risk factors from the size and value decile portfolios used as test assets:
 - Small-minus-big (SMB): the return of a portfolio of small stocks in excess of the return on a portfolio of large stocks
 - High-minus-low (HML): the return of a portfolio of stocks with a high book-to-market ratio (value stocks) in excess of the return on a portfolio of stocks with a low book-to-market ratio (growth stocks)

- Fama and French then showed that the correlation between the market portfolio and the SMB and HML factors is relatively small, which supports their initial hypothesis
- To provide further evidence in support of their hypothesis, Fama and French proposed that we estimate 3 betas for each test asset:
 - A market beta, β^M , from the asset's exposure to the market
 - A size beta, β^{SMB} , from the asset's exposure to the SMB
 - A value beta, β^{HML} , from the asset's exposure to the HML
- To estimate these betas, Fama and French proposed that we run the following regression for each test asset i :

$$r_i(t) - r_F = \alpha_i^{FF} + \beta_i^M (r^M(t) - r_F) + \beta_i^{SMB} r^{SMB}(t) + \beta_i^{HML} r^{HML}(t) + e_i^{FF}(t)$$

- Fama and French then showed that the size and value betas help explain the pattern in the expected returns of the size and value decile portfolios, where the market betas were failing
- The evidence that they provided was so strong that they concluded that the two factors that they uncovered, the size (SMB) and the value (HML) factors must capture some important macroeconomic risk and that all assets must be exposed to these two factors in addition to the market
- The main asset pricing implication of their theory is that the risk premium of any asset i is a combination of the market risk premium, the size and the value premia:

$$E(r_i) - r_F = \beta_i^M E(r^M - r_F) + \beta_i^{SMB} E(r^{SMB}) + \beta_i^{HML} E(r^{HML})$$

- To this day, we don't know for sure what macroeconomic risk do the size and value factors capture
- Fama and French initially hypothesized that the value factor captures *economic distress* in firms
- Their evidence to support this view was not very strong, and later on other researchers showed that this view is not really supported by the data
- Then Obreja (Review of Financial Studies 2013) came and blew everybody out of the water... <https://scholar.google.com/citations?user=8tFnRjMAAAAJ>